

Making of Chameleon Transformation FX

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Figure 1: Chameleon transformation

ABSTRACT

In the world of *Kung Fu Panda 4*, the transformation FX emerges as one of the central narrative elements, showcasing the primary power of the film’s key villain, the Chameleon. This effect imbues the character with a scary, unsettling, ability to morph into different forms, varying significantly in size and shape. It enhances the storytelling by allowing the chameleon to be ubiquitously present in various guises and places. Our FX team embarked on a journey to develop a robust transformation system capable of handling numerous characters. It had to be flexible enough to cater to the needs of a large group of artists and adaptable across a wide range of shots from dynamic kung fu action to close-ups with subtle camera movement. Additionally, this work involved cooperative efforts with multiple departments, such as animation, CFX and lighting, resulting in the creation of cross-department workflows.

KEYWORDS

Transformation effects, procedural animation, 2D fluid simulation

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1 EARLY DEVELOPMENT AND EXPERIMENTS

During the early stages of the film, we collaborated with other departments in development. Multiple concepts were explored, and tools were crafted to support our vision.

1.1 Cell Maker Toolset

To help our asset and surfacing team to create variant versions of the Chameleon for each character she transforms into, we developed an efficient toolset to take the character models and generate scales on their skin. Surfacing artists have the ability to paint and adjust the scales to match artistic requirements. This toolset is distinguished by its real-time feedback, enabling fast artist turnarounds. It facilitates the creation of various representatives of the scales including geometry, displacement maps, and normal maps. It is easy to switch between these items. Additionally, it offers the flexibility to customize cell shapes during generation and to apply weight modifications for added size variation.

1.2 Early Experimentation

In our early experimentation, we delved into various types of transformation including rigid body dynamic based cells transformation and “inky-looking” transformation, among others. These exploratory efforts laid a solid foundation for the FX production that followed. The primary objective was to make this transformation effect both terrifying and awe-inspiring. We also anticipated that this effect would be applied across a substantial number of shots and involve numerous artists.

2 FX PRODUCTION

The culmination of our experimental phase led to the desired visual result, which was then applied to our production shots. The process began with generating mask attributes, timed in accordance with cues from the animation department. These masks were crucial in driving two key processes: the shape blending process and the UV

swirling animation using our 2D fluid simulation system. The resulting animated UV attribute were fed directly into the characters' shaders to swirl the textures on the characters.

2.1 Masks Growth for General Timing

We developed tools to procedurally create mask attributes based on the rough timing that the animator provides. These mask tools allow artists to adjust their timing and transitions. The procedural tools could be driven by painted keyframes or growth algorithms on characters.

2.2 Blending Shapes Between Characters

One of the most challenging aspects was blending shapes between character models transforming from one into another, especially considering their varying differences in size, shape, and mesh topology. To start this process, the animation department provided rough transformation animations. The character models were conformed to each other based on their matching shapes and then blended using the generated masks.

For shots requiring more detail and control, models were segmented into smaller parts to be transformed. Artists could also manually adjust the shapes to ensure a seamless and natural-looking transformation. For example, transformations around the eyes and other fine details were manually adjusted to preserve alignment, shape, and alternate timing during the transitions.

2.3 Particle-Based Custom 2D Fluid Solver

To create the UV swirling effect on character skins, we developed a custom particle-based 2D fluid solver. This solver allowed our artists to simulate particles adhering closely to the character's skin surfaces. Key to this process was generating particles on the character's skin surface and embedding the character's UV information into their point attributes. We executed the generation and simulation of those particles at the character's static (rest) position, ensuring consistent stability irrespective of subsequent iterations in character animation. Our artists could use built-in noise functions to generate the swirling motion of the particles. The villain had an established design motif which was used throughout the film in her environments and clothing. It was based on the Chinese pattern style of Taotie. Our solver was capable of incorporating custom hand drawn curves in this style to influence the particles' velocity.

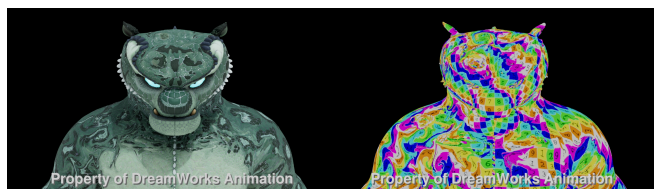


Figure 2: Final render(1); 2D fluid simulation-UV swirling(2)

During the transformation process involving two characters (referred to as “A” and “B”), the solver handled each character differently. For the A character, a mask attribute was used to drive the 2D fluid simulation. Conversely, for the B character, we used a reversed mask attribute to drive its simulation. Post-simulation,

we reversed the timing of B character's simulation back to align with the transformation timing. Artists have the flexibility to easily adjust their timing as needed.

2.4 Material Overrides

The next step involved rasterizing the “uv” attribute from the simulated particles into an OpenVDB. This “uv” OpenVDB was then utilized to override the texture UV coordinates in the characters' shaders, ensuring that the swirling effects were accurately reflected in the final textures. All these processes were executed at the characters' static (rest) positions to maintain consistency and accuracy.

3 MULTIDISCIPLINARY COLLABORATION

The chameleon transformation in *Kung Fu Panda 4* was made possible through extensive collaboration among several departments, each bringing their expertise to the project.

In the film's early stages, we established a close partnership with the surfacing department. We developed the cell maker toolset in collaboration with their technical directors (TDs) who took over for further refinement and iteration.

The transformation FX was heavily reliant on character animation. Our collaboration with the animation team focused on ensuring that the corresponding body parts of the A and B side characters, including arms, legs, fingers, heads, and bodies, were aligned in both camera and 3D world space. This alignment was crucial for the skin blending process. The animators also provided a transparency attribute to indicate the approximate timing of the transitions, which was important for synchronizing our FX efforts.

The FX team handled the transformation of the character's skin geometries. These skin geometries were passed on to the CFX department after the process for clothing adjustment. In cases involving close-fitting clothes or the clothes on nearly static characters, the FX team would take the lead.

The responsibility of final rendering and compositing fell to the lighting department. Given the specific compositing requirements of the transformation FX, we provided rendered layers, masks AOVs and composite treatments in a package to the lighting artists. This package, or “gizmo”, allowed the lighting artists to integrate their character renders seamlessly. We maintained a close collaboration throughout this process to ensure that the final lighting output aligned perfectly with our vision.

4 CONCLUSION

The Chameleon Transformation FX in *Kung Fu Panda 4* was achieved through a fusion of advanced technology and deep cross-departmental collaboration. Our team developed bespoke toolsets, overcame the unique challenges of animating transformations among a large number of characters that vary drastically in size and shape. The collaborative efforts with multiple departments were crucial in achieving a seamless, artistic-driven transformation effect that was integral to the film's narrative. By introducing a unique transformation look, we significantly enhanced the portrayal of the main villain, the Chameleon, making the character more intimidating and impactful in *Kung Fu Panda 4*.